Quantification: 536-63x_Hyp_RD_z08c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	18
interior lacuna count	17
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2672.41	2736	1593.38	px^2
major_axis_length_px	lacuna shape	93.4323	96.5558	38.8621	px
minor_axis_length_px	lacuna shape	39.4424	39.5227	11.9594	px
aspect_ratio	lacuna shape	2.3424	2.126	0.6658	unitless
eccentricity	lacuna shape	0.8783	0.8825	0.0668	unitless
solidity	lacuna shape	0.7925	0.8492	0.1179	unitless
roots_count	headline	10.2941	10	4.9594	count
ring_length_r30_px	headline	321.235	290	166.5	px
ring_length_r60_px	reported	667.353	601	327.779	px
owned_length_px	ownership-dependent	959.858	959.193	754.701	px
edge_count	ownership-dependent	46.1176	42	36.3247	count
mean_edge_length_px	ownership-dependent	20.3984	20.852	3.9712	px
perimeter_px	normalised	286.225	259.557	124.764	px
ring_area_r30_px2	normalised	9646	9821	3107.34	px^2
ring_area_r60_px2	normalised	22305.4	23622	6673.45	px^2
in_frame_fraction_r30	normalised	0.9848	1	0.0472	unitless
in_frame_fraction_r60	normalised	0.9696	1	0.0843	unitless
ring_density_r30	normalised	0.0324076	0.031131	0.00952006	px^-1
ring_density_r60	normalised	0.0290839	0.0278216	0.00813577	px^-1
roots_per_100px_perimeter	normalised	3.678	3.731	0.9025	per 100 px
ring_attached_length_r30_px	network v2	288.471	261	161.234	px
ring_attached_length_r60_px	network v2	519.353	457	295.105	px
ring_length_w_r30_px	network v2	363.109	336.463	186.051	px
ring_length_w_r60_px	network v2	759.272	682.541	366.919	px
sholl_crossings_r10	network v2	7.8235	8	3.3211	count
sholl_crossings_r20	network v2	8.2941	9	4.0738	count
sholl_crossings_r30	network v2	8.9412	8	4.0846	count
loop_count	loops, report only	2.6471	2	3.9042	count
loop_edge_share	loops, report only	0.1367	0.1522	0.1293	share
ring_width_mean_r30_px	width	7.5238	7.3803	0.6293	px

Per field
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measure	value	units
analysed_area_px2	1.00286e+06	px^2
total_skeleton_length_px	26098	px
canalicular_length_density_per_px	0.0260234	px^-1
junction_count	500	count
junction_density_per_px2	0.00049857	px^-2
lacuna_count	18	count

lacunae_per_px2	1.795e-05	px^-2
skeleton_component_count	357	count
mean_component_length_px	73.1036	px
median_component_length_px	24	px
field_length_density_w_per_px	0.0295721	px^-1
field_density_without_flagged_per_px	0.0272192	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	321	px
unattached_component_count	18	count
canalicular_length_density_attached_per_px	0.0260234	px^-1
field_length_density_with_unattached_per_px	0.0263435	px^-1
loop_count	55	count
loop_edge_share	0.1586	share
edge_count	1343	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7749	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25895	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	4713	113.091	66.6058	1.6979	0.8082	0.6869	0.0149	166.949	811.934	False
2	2838	120.116	56.4992	2.126	0.8825	0.5006	-0.0922	183.402	482.588	False
3	593	40.0782	21.7723	1.8408	0.8396	0.7583	0.4372	249.45	834.877	False
4	5444	169.703	43.0843	3.9389	0.9672	0.8492	-0.1663	315.438	989.012	False
5	472	35.2068	21.2692	1.6553	0.7969	0.7528	-0.5629	318.485	581.102	False
6	5032	145.049	50.087	2.8959	0.9385	0.6975	-0.3567	389.803	259.63	False
7	1325	56.519	31.2238	1.8101	0.8335	0.9138	-0.4181	431.483	840.339	False
8	1869	63.382	43.4291	1.4594	0.7284	0.783	-0.4231	489.012	823.251	False
9	4023	130.997	44.5634	2.9396	0.9404	0.8529	-0.2947	478.273	373.626	False
10	2063	91.1753	31.0575	2.9357	0.9402	0.8892	-0.3885	511.078	505.802	False
11	2736	102.242	36.178	2.8261	0.9353	0.8653	-0.4465	632.235	931.07	False
12	2738	96.5558	37.4594	2.5776	0.9217	0.8784	-0.3793	679.917	399.446	False
13	3288	105.667	42.059	2.5123	0.9174	0.8761	-0.3815	724.683	177.336	False
14	797	46.6745	23.1748	2.014	0.868	0.8935	-0.2375	760.548	828.605	False
15	1183	68.5981	39.5227	1.7357	0.8173	0.5796	-0.4587	845.742	872.086	False
16	4334	128.945	44.2583	2.9135	0.9393	0.8796	-0.1396	903.815	374.562	False
17	1983	74.3475	38.2775	1.9423	0.8573	0.8154	-0.0273	970.974	775.836	False
18								7.45	906.793	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	19	469	863	959.193	46	20.852	7.4871

2	19	723	1431	1324.53	81	16.3522	6.6187
3	6	202	455	783.482	32	24.4838	7.3169
4	10	325	580	1119.45	52	21.5278	7.3803
5	5	273	610	1764.85	85	20.7629	6.8543
6	20	632	1263	3256.5	156	20.875	7.0906
7	4	90	197	79.1838	7	11.312	7.454
8	8	165	372	227.338	12	18.9448	6.9494
9	13	411	890	1092.82	61	17.9151	8.8041
10	10	286	601	967.637	42	23.039	8.2735
11	6	200	468	298.563	17	17.5626	7.9746
12	11	305	670	1297.54	48	27.0322	8.1069
13	10	349	860	1189.62	50	23.7924	7.3406
14	7	150	315	327.948	15	21.8632	7.6486
15	11	290	462	470.617	31	15.1812	7.1659
16	10	375	855	842.269	33	25.5233	8.5809
17	6	216	453	316.049	16	19.753	6.858
18	4	117	210	236.208	9	26.2454	7.4944

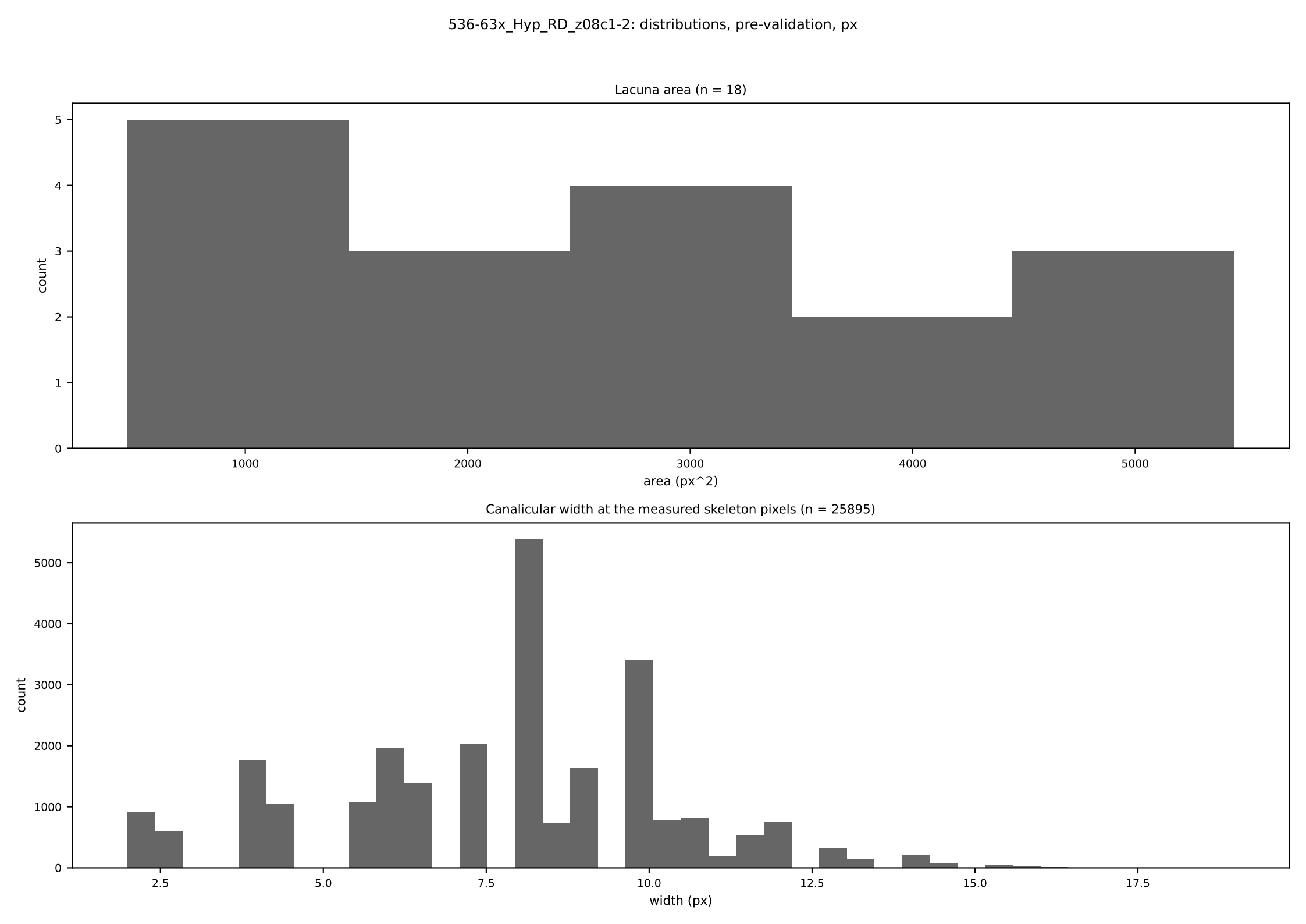
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	442.446	12800	28014	1	1	0.0366406	0.030806	4.2943
2	514.931	14099	31670	1	1	0.0512802	0.0451847	3.6898
3	128.539	4684	10910	1	1	0.0431255	0.0417049	4.6678
4	423.747	11743	23622	0.8152	0.7003	0.0276761	0.0245534	2.3599
5	117.539	5781	16713	1	1	0.0472237	0.0364985	4.2539
6	487.973	14467	32198	1	1	0.0436856	0.039226	4.0986
7	156.16	5263	12719	1	1	0.0171005	0.0154886	2.5615
8	228.202	6707	15935	1	1	0.0246012	0.0233448	3.5057
9	348.433	12410	27080	1	1	0.0331185	0.0328656	3.731
10	227.38	9187	22623	1	1	0.031131	0.0265659	4.3979
11	274.735	10452	25593	1	1	0.0191351	0.0182863	2.1839
12	250.693	9821	24082	1	1	0.0310559	0.0278216	4.3878
13	298.534	10681	26849	1	1	0.0326748	0.032031	3.3497
14	131.776	6343	14155	1	1	0.0236481	0.0222536	5.3121
15	259.557	9112	19798	1	1	0.0318262	0.0233357	4.238
16	340.877	12243	29334	1	0.982	0.0306298	0.0291471	2.9336
17	234.309	8189	17896	0.9258	0.8017	0.0263769	0.0253129	2.5607
18	69.2132	2839	8492	0.5962	0.5598	0.0412117	0.0247292	5.7792

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	452	769	531.375	988.335	11	10	13
2	694	1052	813.127	1610.74	14	11	16
3	202	435	228.924	526.002	6	3	5
4	325	552	368.291	663.198	9	10	7
5	216	457	316.492	708.583	6	6	9
6	563	1136	709.985	1430.87	13	16	16
7	76	76	102.326	221.995	4	0	2
8	150	205	184.196	422.818	3	5	5
9	371	812	458.635	1009.29	8	11	13
10	243	396	326.735	682.541	12	7	10
11	149	242	228.865	539.742	6	9	7
12	305	602	350.978	779.181	8	9	8
13	261	536	373.551	951.796	8	11	11
14	150	263	174.439	363.605	4	5	5

15	273	427	336.463	534.529	7	7	8
16	329	642	423.019	955.595	10	15	12
17	145	227	245.451	518.801	4	6	5
18	117	184	132.569	246.451	2	3	2


536-63x_Hyp_RD_z08c1-2: distributions, pre-validation, px
Lacuna area (n = 18)
count
area (px^2)
0
1
2
3
4
5
1000
2000
3000
4000
5000
Canalicular width at the measured skeleton pixels (n = 25895)
count
width (px)
0
1000
2000
3000
4000
5000
2.5
5.0
7.5
10.0
12.5
15.0
17.5